Kappa HCE Loss — Complete Mathematics

H+c ($H \to WW$) analysis

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Abstract

The **kappa hierarchical cross-entropy (kappa HCE)** loss is the current best loss for the $H \to WW$ $H + c$ MVA (v32: `hplusc_vs_all` AUC **0.975**). It replaces manually-defined hierarchical groups with *dynamic* groups derived from the geometry of the classifier's own output layer, and couples them to a differentiable significance term. This note derives all of the mathematics.

1 Setup and notation

Symbol	Meaning
C	number of classes (13 for $H \to WW$)
s	signal class index (hplusc)
$z \in \mathbb{R}^C$	logits from the network for one event
$p_i = \mathrm{softmax}(z)_i$	predicted posterior $P(\text{class } i \mid x)$, $\sum_i p_i = 1$
$W \in \mathbb{R}^{C\times h}$	weight matrix of the final linear layer ($h = 32$)
$W_j \in \mathbb{R}^h$	row j of W — the learned *template vector* for class j
y	integer truth label of the event
N_j	number of training events in class j
τ	temperature for soft group membership
λ	weight of the significance term (`lam`)
k_{fine}	weight of the fine-separation term

The logit for class j is $z_j = W_j \cdot \phi(x) + b_j$, where $\phi(x)$ is the penultimate activation. So **each row W_j is the direction in feature space that the network associates with class j.** Two classes whose templates point in similar directions are "confusable" — this is the geometric intuition the loss exploits.

2 Dynamic groups from cosine similarity

Define the cosine similarity between the signal template and every class template:

$$c_j \;=\; \cos\bigl(W_s, W_j\bigr) \;=\; \frac{W_s \cdot W_j}{\|W_s\|\,\|W_j\|}, \qquad c_s = 1. \tag{1}$$

This is recomputed **every forward pass** from the live weights, so the grouping evolves as the model trains.

2.1 Hard split (used by the group term)

$$\mathcal{P} = \{\, j : c_j \geq 0 \,\} \quad \text{(signal-like, includes } s\text{)}, \qquad \mathcal{N} = \{\, j : c_j < 0 \,\}, \qquad n_{\mathcal{N}} = |\mathcal{N}|. \tag{2}$$

2.2 Soft membership (used by the fine and significance terms)

$$\alpha_j \;=\; \sigma\!\left(\frac{c_j}{\tau}\right) \;=\; \frac{1}{1 + e^{-c_j/\tau}} \;\in (0,1). \tag{3}$$

α_j is the **soft "kappa"** — the degree to which class j is treated as belonging to the signal-like fine group. τ controls sharpness: at $\tau = 0.3$, $\alpha > 0.5$ once $c_j > 0$ and $\alpha < 0.05$ by $c_j < -0.6$. Smaller τ gives harder (step-like) membership.

Stop-gradient

In code α is computed from a *detached* c_j, so α acts as a read-out gate, not a free parameter driven by the loss. The template geometry (c_j) is shaped **only** by the CE terms below; the significance term cannot push α around. This is precisely why the kappa HCE loss does **not** suffer the kappa-collapse ($\to 0$) or kappa-convergence ($\to 1$) pathologies of the earlier learned-kappa losses.

3 The three loss terms

$$\boxed{\mathcal{L} \;=\; \mathcal{L}_{\text{group}} \;+\; k_{\text{fine}}\,\mathcal{L}_{\text{fine}} \;+\; \lambda\,\mathcal{L}_{\text{sig}}}$$

3.1 Group term — hard-split coarse CE

Collapse the $\mathcal{P}$ classes into a single merged column and keep each $\mathcal{N}$ class separate. Grouped probabilities $q \in \mathbb{R}^{n_{\mathcal{N}}+1}$:

$$q_k = p_{\mathcal{N}_k} \;(k < n_{\mathcal{N}}), \qquad q_{n_{\mathcal{N}}} = \sum_{j \in \mathcal{P}} p_j. \tag{4}$$

Map each event's truth to a group index $g(y)$: its own column if $y \in \mathcal{N}$, else the merged column $n_{\mathcal{N}}$. Then

$$\mathcal{L}_{\text{group}} \;=\; -\,\frac{\sum_x w_{g(y)}\,\log q_{g(y)}(x)}{\sum_x w_{g(y)}}, \tag{5}$$

a weighted NLL. The group weights are inverse-frequency:

$$w_k = \widehat{\text{invfreq}}(\mathcal{N}_k) \;(k < n_{\mathcal{N}}), \qquad w_{n_{\mathcal{N}}} = \frac{1}{\sum_{j \in \mathcal{P}} N_j}, \tag{6}$$

where $\widehat{\text{invfreq}}(j) = \dfrac{\sum_i N_i}{N_j}$ normalised so $\max_j = 1$.

Why merge the positives? A merged column contributes no term that pushes the $\mathcal{P}$ templates *apart*. If each positive class had its own softmax column, the CE gradient would repel their weight vectors, destabilising the very cosine structure that defines the groups. Merging keeps the signal-like cluster cohesive.

3.2 Fine term — soft-membership conditional CE

Build an α-gated conditional distribution over classes:

$$\tilde{p}_i(x) \;=\; \frac{\alpha_i\, p_i(x)}{\sum_j \alpha_j\, p_j(x)}. \tag{7}$$

Per-event fine loss (inverse-frequency weighted NLL), then each event weighted by α of its own true class:

$$\ell_{\text{fine}}(x) = -\,\widehat{\text{invfreq}}(y)\,\log \tilde{p}_y(x), \qquad \mathcal{L}_{\text{fine}} \;=\; \frac{\sum_x \alpha_{y(x)}\, \ell_{\text{fine}}(x)}{\sum_x \alpha_{y(x)}}. \tag{8}$$

The double appearance of α — inside $\tilde{p}$ (softens the denominator to the signal-like region) and as the per-event weight $\alpha_{y(x)}$ (down-weights background events) — is what makes this term focus purely on separating the signal from its near neighbours. Its final contribution is scaled by k_{fine}.

3.3 Significance term — kappa discriminant

Define the **kappa discriminant** with α playing the role of κ:

$$D(x) \;=\; \frac{p_s(x)}{p_s(x) + \displaystyle\sum_{j\neq s} \alpha_j\, p_j(x)} \;\in (0,1]. \tag{9}$$

Signal-like classes ($\alpha_j \to 1$) stay in the denominator and **dilute** D — so D only grows when p_s genuinely dominates its near neighbours. Background-like classes ($\alpha_j \to 0$) drop out and cannot inflate the denominator. The term is a **batch-mean** on each half (composition-invariant, spike-free):

$$\mathcal{L}_{\text{sig}} \;=\; \underbrace{\frac{1}{n_{\text{sig}}} \sum_{x:\, y=s} -\log D(x)}_{\text{push } D\uparrow \text{ for signal}} \;+\; \underbrace{\frac{1}{n_{\text{bkg}}} \sum_{x:\, y\neq s} \log\bigl(1 + D(x)\bigr)}_{\text{push } D\downarrow \text{ for background}}. \tag{10}$$

- $-\log D$ diverges as $D \to 0$: strong pull to raise p_s on true signal.
- $\log(1 + D)$ is bounded and floored at 0 (unlike the old $\log B$ which ran to $-\infty$): it gently suppresses p_s on backgrounds without dominating.
- Both gradients act on p (via z), aligned with the CE terms — no probability-suppression artefact.

The $\frac{1}{n_{\text{sig}}}, \frac{1}{n_{\text{bkg}}}$ normalisation (with a guard for signal-empty batches) removes the batch-composition dependence that made the older summed significance losses spike on signal-heavy batches.

4 Why it works: the natural curriculum

The grouping is emergent, so training passes through phases automatically:

1. **Early (random W):** cosines are scattered around 0; many classes land in $\mathcal{P}$. $\mathcal{L}_{\text{fine}}$ is effectively a broad multiclass CE over most of the 13 classes $\to$ the model first learns coarse structure.
2. **Middle:** $\mathcal{L}_{\text{group}}$ drives dissimilar backgrounds' templates to anti-align ($c_j < 0$), so they leave $\mathcal{P}$. The fine term's support shrinks toward the signal-like cluster.
3. **Late:** only the genuinely signal-like classes remain positive. In v32 by epoch 49 that is just **hplusb, ggH, VBF**. $\mathcal{L}_{\text{fine}}$ has converged to a near-binary hplusc-vs-near-neighbour problem.

So a *single* objective interpolates from coarse multiclass to fine binary discrimination without any manual schedule.

4.1 v32 learned membership (from `best_model.pt`, $\tau = 0.3$)

Class	c_j	α_j	Group
H+c	$+1.000$	0.966	signal
H+b	$+0.355$	0.765	$\mathcal{P}$
VBF	$+0.277$	0.716	$\mathcal{P}$
ggH	$+0.097$	0.580	$\mathcal{P}$
V+Jets	-0.244	0.307	$\mathcal{N}$
ttHtoBB	-0.434	0.191	$\mathcal{N}$
ZH	-0.478	0.169	$\mathcal{N}$
ttHnonBB	-0.498	0.160	$\mathcal{N}$
ggZH	-0.522	0.149	$\mathcal{N}$
tt	-0.621	0.112	$\mathcal{N}$
Diboson	-0.670	0.097	$\mathcal{N}$
WH	-0.708	0.086	$\mathcal{N}$
ST	-0.733	0.080	$\mathcal{N}$

The model discovers, without being told, that H+b / VBF / ggH are the charm-like Higgs backgrounds that matter, and rejects everything else.

5 Inference / post-training discriminant

At inference the trained templates give fixed κ_j (via α_j or a separately-optimised vector). The event score is the same discriminant:

$$D_{\text{sig}}(x) = \frac{p_s(x)}{p_s(x) + \sum_{j \neq s} \kappa_j \, p_j(x)}. \tag{11}$$

Two ways to set κ:

- **From the model:** $\kappa_j = \alpha_j = \sigma(c_j/\tau)$ (or the smooth mapping $(1+c_j)/2$, or clamp $\max(0, c_j)$).
- **Optimised per working point:** maximise $S/\sqrt{B}$ at a target signal efficiency via differential evolution, $\kappa_j \in [0, 5]$. For v32/v20 this gives only $\sim +1.2\%$ over uniform κ, because the trained posteriors already encode the structure.

$S/\sqrt{B}$ at a working point uses **raw event counts** (no inverse-frequency): $S = \sum_{x \in \text{sig}} ⊮[D(x) > t]$, $B = \sum_{x \in \text{bkg}} ⊮[D(x) > t]$, with threshold t set to the target signal efficiency.

6 Relation to earlier losses

Loss	**Groups**	κ **source**	**Failure mode**	**AUC**
Flat CE (v10)	none	—	13-class imbalance collapse	0.51
Hierarchical CE (v12–v15)	manual	—	needs hand-tuned groups	≤ 0.965
$-S/\sqrt{B}$ (v18s)	manual	learned, clamp	κ collapse $\to 0$	0.928
$\log B$ (v18/19logB)	manual	learned, smooth	κ converge $\to 1$	0.961
Discriminant-aware (v20)	manual	learned, smooth	— (best fixed-group)	0.973
Kappa HCE (v32)	**dynamic**	$\alpha = \sigma(c/\tau)$, **detached**	—	**0.975**

The kappa HCE loss keeps v20's discriminant-aware significance term but (i) replaces manual groups with emergent cosine groups and (ii) detaches κ so the significance objective can never destabilise the grouping — giving the best AUC and a hyperparameter-light, self-scheduling training.